

OneGrow: Unified Temporal Plant Image and Mask Generation

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Abstract. Image-based crop phenotyping benefits from image series that capture plant development together with organ-level labels, yet such paired data are scarce. We present *OneGrow*, a latent flow-matching model that jointly models wheat images and their organ-segmentation masks over time. One frozen image autoencoder encodes both modalities, and a reveal specifies which content is observed context, so one trained model covers mask-to-image synthesis, image-to-mask segmentation, temporal forecasting, and spatial completion. A sliding-window roll-out extends generation beyond the training horizon by revealing already-generated frames as context. We train jointly on a single-frame and a multi-year temporal wheat dataset using pseudo-labels from a pre-trained segmentation model. We evaluate segmentation and image quality against held-out references, and assess the multi-task and temporal behavior qualitatively.

Keywords: plant phenotyping · flow matching · image generation · wheat · time-series

1 Introduction

Image-based phenotyping derives plant traits from field imagery and is widely used to study crop development [13, 17, 20]. Organ-level structure comes from segmentation masks, whose annotation is labor-intensive, so labeled data cover only a small fraction of collected imagery [20]. Temporal development requires repeat imaging across a season, which few platforms undertake [17], so imagery providing both remains scarce.

Generative models may expand such data, but existing plant models either predict imagery at unobserved time points without producing structural labels [4, 5, 19] or generate labeled image-mask pairs for a single frame [7]. Unified generative models let each input and output act as context or target [3, 11], yielding joint image and segmentation generation [15, 21, 23]. These target general scenes and dense video, not sparse seasonal observations.

We present *OneGrow*, a latent flow-matching model of wheat images and their organ masks over time (Fig. 1).

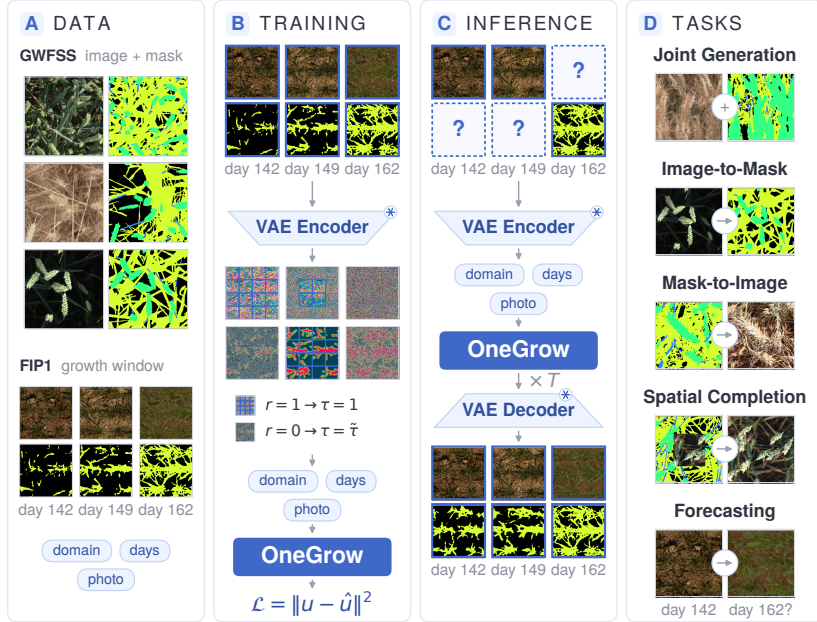


Fig. 1: Overview of OneGrow. **(A)** Training pairs every image with a mask, from single-frame GWFSS samples and multi-frame FIP1 windows, with the conditioning metadata. **(B)** One frozen autoencoder encodes the image and mask views of a window, the reveal marks each patch as clean context or generated, and one transformer regresses the joint velocity field. **(C)** At inference, revealed patches stay clean and the rest are generated from noise, then decoded. **(D)** One model, many tasks, each selected by a different reveal.

2 Method

Shared image-mask representation. Images and masks are encoded by the same frozen, pretrained image variational autoencoder (VAE) [10, 16]. To encode a per-pixel class map with an image encoder, we *color-code* it into an RGB image through a fixed palette [15]. The palette colors stay separable through the round-trip, so assigning each decoded pixel to its nearest palette color recovers the mask at 0.990 mean intersection-over-union (mIoU), and masks and images differ only in a *modality* index.

Reveal-based flow matching. A growth window is a collection of *views*, one per modality and frame, each split into patches. The binary reveal r marks every patch as observed context or as content to generate, and it is what selects the task (Fig. 1D). Recording it per patch rather than per view lets spatial completion observe part of an image and generate the rest. OneGrow is a flow-matching diffusion transformer (DiT) [12, 14] that connects a clean latent z

and Gaussian noise ϵ with the interpolant $z_\tau = \tau z + (1 - \tau)\epsilon$, regressing the velocity target $u = z - \epsilon$, where $\tau=1$ is clean data and $\tau=0$ is noise. The reveal gives each patch one of only two flow times, $\tau=1$ on revealed patches and a shared $\tilde{\tau}$ on generated ones, drawn per window from a logit-normal schedule [6]. Register tokens carry the acquisition domain and photometric statistics, dropped for classifier-free guidance [8].

Temporal roll-out. A sequence can be longer than the training window, so we roll the model out autoregressively with a stride-one sliding window [3]. Each step reveals the already-generated frames of the overlap as context and generates the next frame with its own mask, so the sequence stays consistent.

3 Data and Training

The *GWFS* dataset [20] provides about 64k unlabeled images from nine institutions, without temporal structure. We pseudo-label them with the winning entry of its segmentation challenge [2] and train on the resulting pairs [22], holding out its 1,096-image human-annotated test set. That entry was ranked on this set but never trained on it, so the evaluation stays independent of the teacher. The *FIP1* dataset [17] follows about 4,000 winter-wheat plots over six years, and we form each plot sequence into four-frame windows. One model is trained jointly on both with the frozen SD3.5 VAE [6].

4 Results

Segmentation and mask adherence. On the held-out human-annotated GWFS masks, OneGrow reaches an mIoU of 0.697 against 0.740 for the challenge winner [2] (Tab. 1). The gap is largest on the thin stem class, which is also the winner’s weakest at 0.488. The winner produced the pseudo-labels OneGrow trains on, so its accuracy bounds what OneGrow can learn. We further measure *mask adherence*, how well a generated image respects the organ layout it was conditioned on. The winner re-segments the generated image, and that segmentation is scored against the conditioning mask. Adherence reaches 0.749, on par with the 0.740 the winner reaches on real images. The teacher and the scorer are the same model, so the value may favor layouts it segments reliably.

Mask-to-image quality. We compare against Mask-NN, a training-free baseline returning the training image whose mask latent is nearest. OneGrow attains a lower paired Learned Perceptual Image Patch Similarity (LPIPS) [24] than Mask-NN, 0.426 against 0.548, while Mask-NN attains a lower Fréchet distance in DINOv2 feature space (FD-DINOv2) [18], 257.0 against 322.2, and a lower CLIP maximum mean discrepancy (CMMD) [9], 16.0 against 19.3. Mask-NN returns real training images, so it is favored on the two set-level metrics, but it cannot return the particular held-out frame that paired LPIPS scores.

Table 1: Image-to-mask segmentation and mask adherence (third GWFSS row, by cycle consistency), per-organ and mean intersection-over-union, higher is better. The GWFSS reference is the held-out human-annotated masks, and FIP1’s is the teacher pseudo-labels.

Dataset	Method	Head	Stem	Leaf	Background	mIoU $\uparrow$
GWFSS	Challenge winner [2]	0.847	0.488	0.833	0.791	0.740
	OneGrow (image-to-mask)	0.811	0.412	0.805	0.758	0.697
	OneGrow (mask-to-image-to-mask)	0.838	0.504	0.845	0.807	0.749
FIP1	OneGrow (image-to-mask)	0.701	0.252	0.803	0.843	0.650

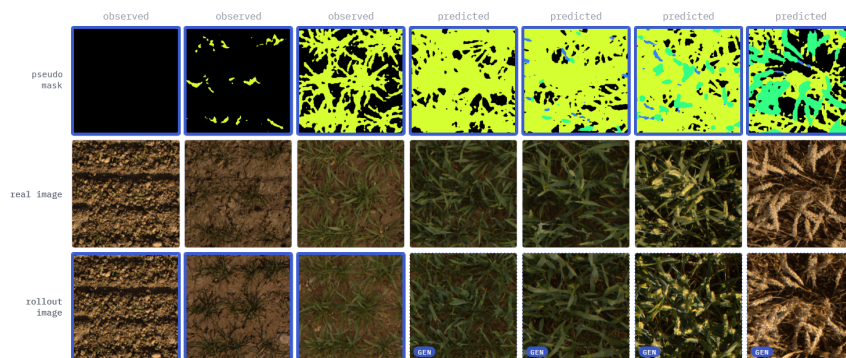


Fig. 2: Temporal roll-out on a FIP1 growth sequence, one column per time step. *Top:* the teacher pseudo-mask, revealed for every frame. *Middle:* the real image, for reference. *Bottom:* the roll-out, seeded by three real frames (blue borders), each later frame (GEN) generated from its mask and the three preceding frames. Masks: background (black), leaf (yellow-green), stem (blue), head (green).

Multi-task generation and temporal roll-out. The same trained model performs every task of Fig. 1D. Rolled out over a FIP1 sequence, it stays consistent well beyond the four-frame training window (Fig. 2). Further qualitative results and implementation details are in the supplementary material.

5 Conclusion

OneGrow covers several image-mask tasks in one trained model, and its sliding-window roll-out conditions each new image on its mask and preceding frames. The pseudo-labels inherit the teacher’s biases, training covers wheat only, and the mask-conditioned time-series generation is so far shown qualitatively. The roll-out could render plant-model organ mask sequences [1] as imagery.

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